

EXPERIMENT 2: INPUT & OUTPUT CHARACTERISTICS OF A BJT IN CE, CB, AND CC CONFIGURATIONS

1. Aim

- To plot the input and output characteristics of a BJT in the Common Emitter (CE) configuration, and to determine the current gain β , input resistance r_i , and output resistance r_o .
- To plot the input and output characteristics of a BJT in the Common Base (CB) configuration, and to determine the current gain α , input resistance r_i , and output resistance r_o .
- To plot the input and output characteristics of a BJT in the Common Collector (CC) configuration, and to determine the current gain γ , input resistance r_i , and output resistance r_o .
- To verify the relationships $\beta = \alpha/(1-\alpha)$ and $\gamma = \beta + 1$ from the measured data.

2. Apparatus / Components Required

S.No.	Component / Equipment	Specification	Quantity
1	BC547 transistor	NPN, TO-92 package	1
2	Regulated DC power supply	0–30 V, dual output	2
3	Milliammeter	0–10 mA and 0–30 mA range	2
4	Microammeter	0–500 μ A range	1
5	Voltmeter	0–1 V and 0–30 V range	2
6	Fixed resistors	1 k Ω , 4.7 k Ω , 10 k Ω	as needed
7	Breadboard, connecting wires, probes	—	as needed

3. Theory

3.1 BJT Fundamentals

A Bipolar Junction Transistor (BJT) is a three-terminal device — emitter (E), base (B), and collector (C) — formed by two back-to-back PN junctions. In normal (active-region) operation, the base–emitter junction is forward-biased and the base–collector junction is reverse-biased. A small base current controls a much larger collector current, giving current gain. Depending on which terminal is common to both the input and output loops, three configurations are possible: Common Emitter (CE), Common Base (CB), and Common Collector (CC), each with its own characteristic input and output curves and current-gain definition.

3.2 Common Emitter (CE) Configuration

In the CE configuration, the emitter is common to both input (base–emitter) and output (collector–emitter) loops. The input characteristic (I_B vs V_{BE} at constant V_{CE}) resembles a diode's forward characteristic. The output characteristic (I_C vs V_{CE} at constant I_B) shows a steep rise in the saturation region followed by a nearly flat active region, whose slight upward slope reflects a finite output resistance. The current gain $\beta = \Delta I_C / \Delta I_B$ is typically 50–300.

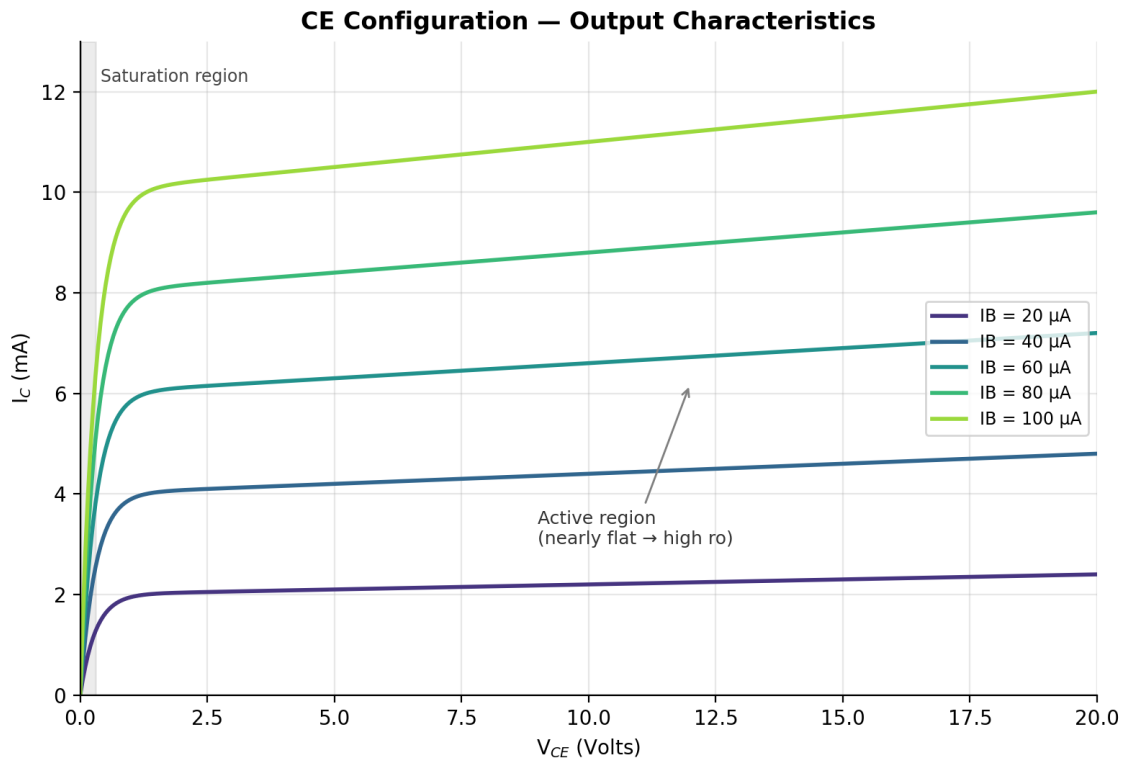


Fig. 1: Typical CE output characteristics — I_C vs V_{CE} for several fixed values of I_B .

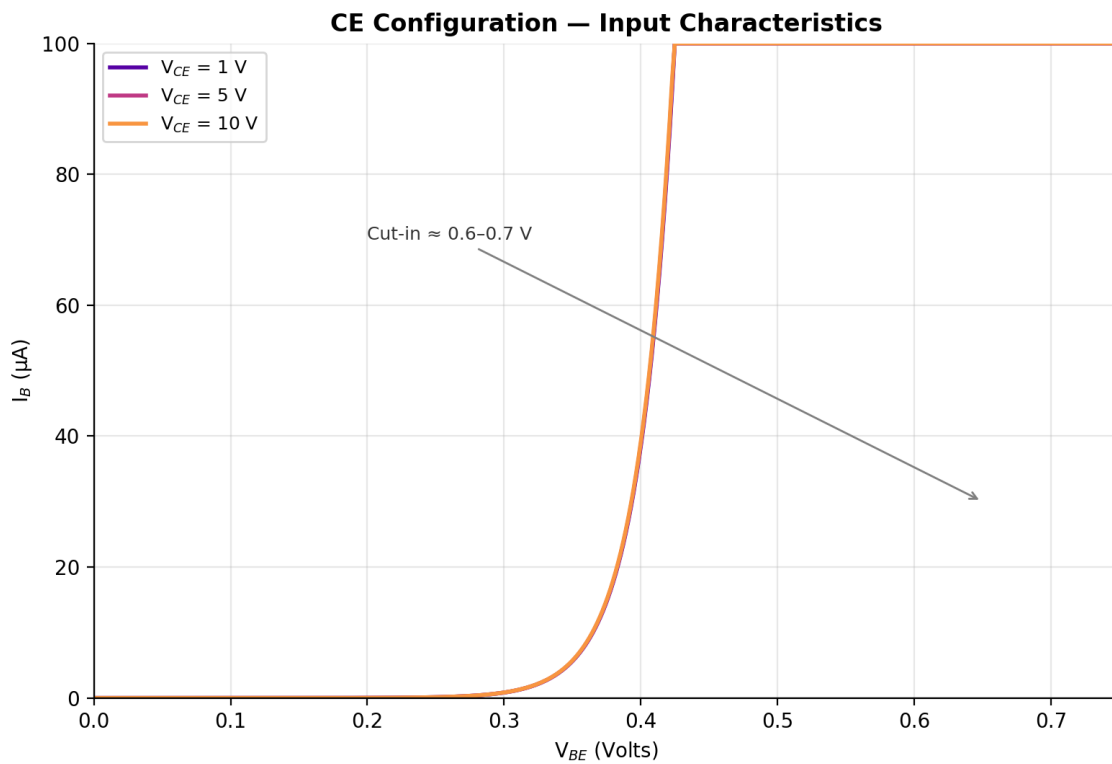


Fig. 2: Typical CE input characteristics — I_B vs V_{BE} for several fixed values of V_{CE} .

3.3 Common Base (CB) Configuration

In the CB configuration, the base is common to both loops. The input characteristic (I_E vs V_{EB} at constant V_{CB}) is again diode-like. The output characteristic (I_C vs V_{CB} at constant I_E) is extremely flat, since the collector current is set almost entirely by the emitter current and is nearly independent of V_{CB} — giving CB circuits a very high output resistance. The current gain $\alpha = \Delta I_C / \Delta I_E$ is always slightly less than 1 (typically 0.95–0.99).

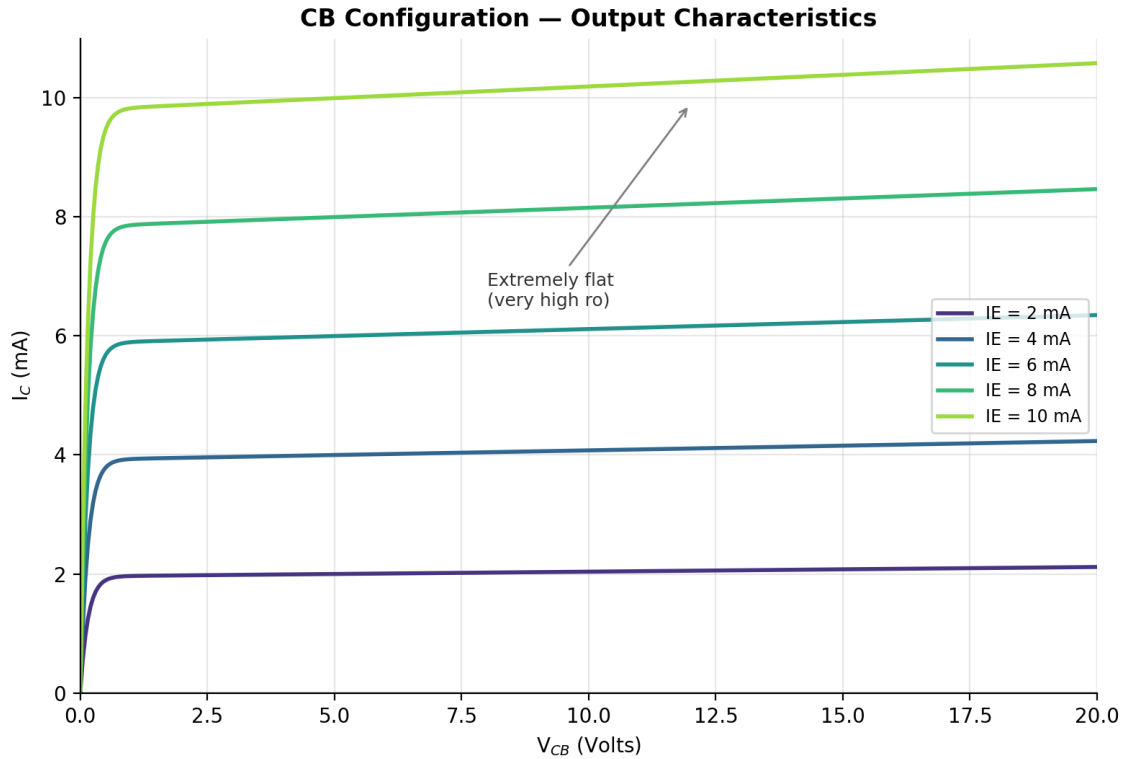


Fig. 3: Typical CB output characteristics — I_C vs V_{CB} for several fixed values of I_E .

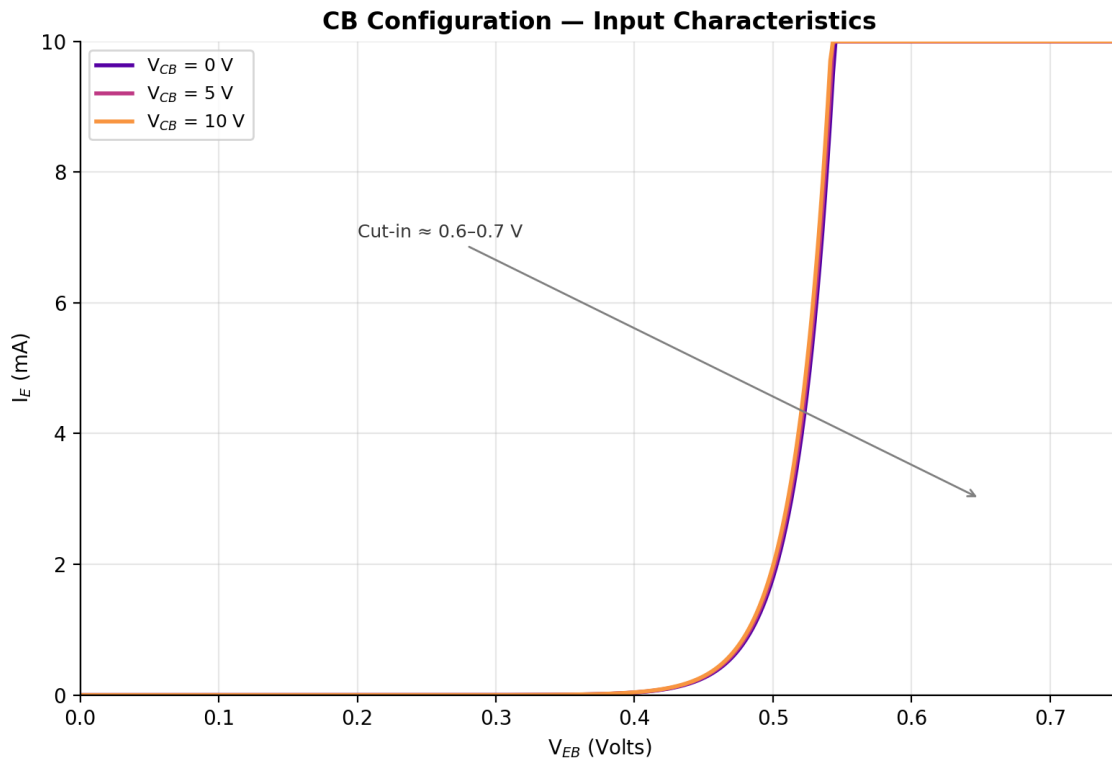


Fig. 4: Typical CB input characteristics — I_E vs V_{EB} for several fixed values of V_{CB} .

3.4 Common Collector (CC) Configuration

In the CC configuration (also called an emitter follower), the collector is common to both loops. The output characteristic (I_E vs V_{EC} at constant I_B) closely resembles the CE output characteristic, since $I_E \approx I_C$. The input characteristic is normally plotted as I_B vs V_{BC} at constant V_{EC} , and falls off as the base–collector reverse bias increases. The current gain $\gamma = \Delta I_E / \Delta I_B \approx \beta + 1$, so CC circuits provide current gain close to that of the CE configuration but are used mainly for their high input resistance and low output resistance (voltage buffering).

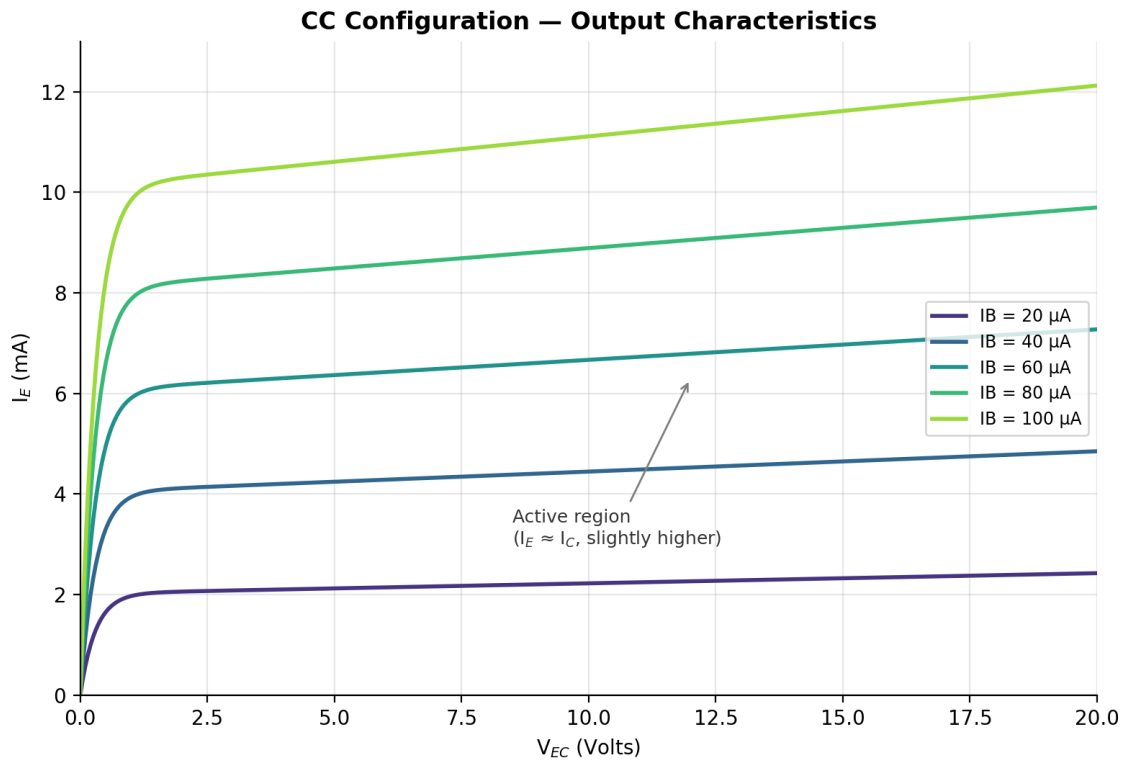


Fig. 5: Typical CC output characteristics — I_E vs V_{EC} for several fixed values of I_B .

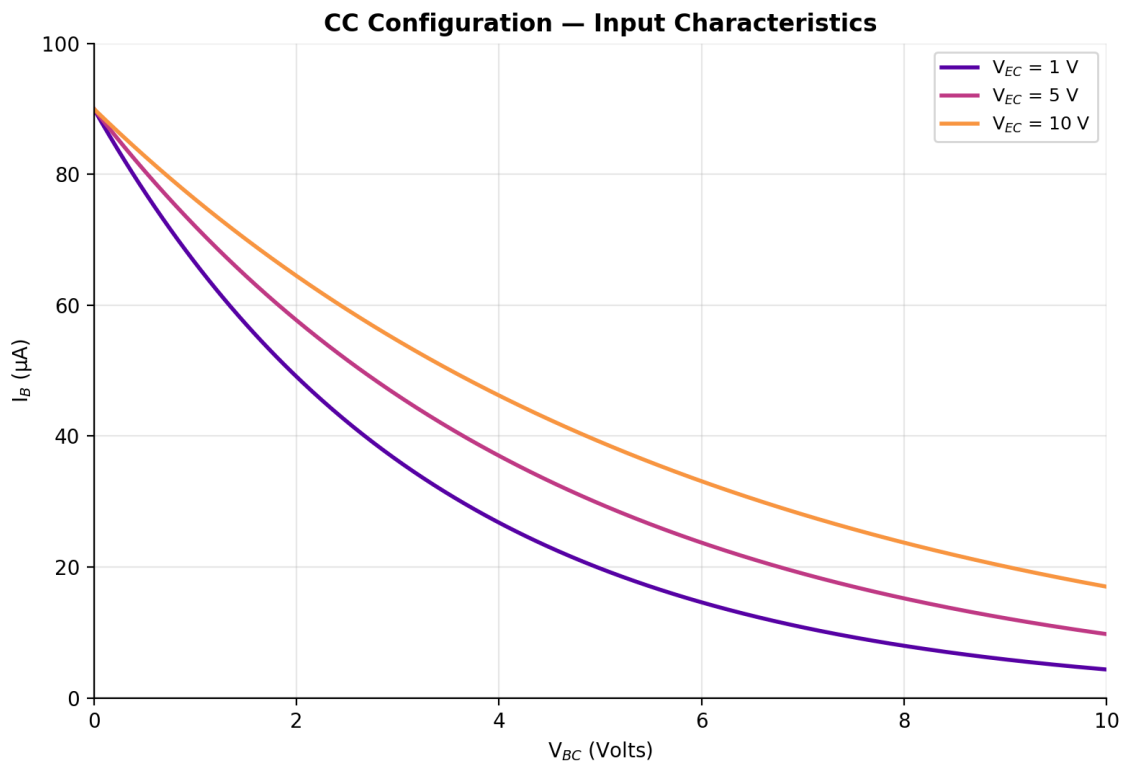


Fig. 6: Typical CC input characteristics — I_B vs V_{BC} for several fixed values of V_{EC} .

4. Part A — CE Configuration: Output Characteristics

Circuit Connections

1. Connect the transistor in CE configuration: emitter grounded (common), base current supplied through a 10 k Ω resistor from the base supply with a microammeter (0–500 μ A) in series, and collector connected through a milliammeter (0–10 mA) to the collector supply.
2. Connect a voltmeter (0–1 V) across base–emitter and a voltmeter (0–30 V) across collector–emitter.

Procedure

3. Set the base supply to give $I_B = 20 \mu\text{A}$ (adjust and confirm on the microammeter) and keep it fixed for this run.
4. Starting with $V_{CE} = 0 \text{ V}$, increase V_{CE} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_C at each step.
5. Repeat steps 1–2 for $I_B = 40, 60, 80, \text{ and } 100 \mu\text{A}$, tabulating I_C vs V_{CE} for each.
6. Switch off the supply. Plot I_C (Y-axis) vs V_{CE} (X-axis) as a family of curves, one for each I_B .

Observation Table — CE Output Characteristics

S.No.	VCE (V)	IC (mA) at $I_B=20\mu\text{A}$	IC (mA) at $I_B=60\mu\text{A}$	IC (mA) at $I_B=100\mu\text{A}$
1				
2				
3				
4				
5				
6				
7				
8				

5. Part B — CE Configuration: Input Characteristics

Circuit Connections

7. Retain the CE circuit of Part A.

Procedure

8. Set $V_{CE} = 1 \text{ V}$ and hold it fixed for this run.
9. Increase V_{BE} in steps of 0.05 V from 0 to 0.4 V, then in steps of 0.02 V up to 0.7 V, recording I_B at each step.
10. Repeat step 2 for $V_{CE} = 5 \text{ V}$ and $V_{CE} = 10 \text{ V}$.
11. Switch off the supply. Plot I_B (Y-axis) vs V_{BE} (X-axis) as a family of curves, one for each V_{CE} .

Observation Table — CE Input Characteristics

S.No.	VBE (V)	IB (μ A) at $V_{CE}=1\text{V}$	IB (μ A) at $V_{CE}=5\text{V}$	IB (μ A) at $V_{CE}=10\text{V}$
1				

2				
3				
4				
5				
6				
7				
8				

6. Part C — CB Configuration: Output Characteristics

Circuit Connections

12. Connect the transistor in CB configuration: base grounded (common), emitter current supplied through a 1 k Ω resistor from the emitter supply with a milliammeter (0–10 mA) in series, and collector connected through a milliammeter (0–10 mA) to the collector supply.
13. Connect a voltmeter (0–1 V) across emitter–base and a voltmeter (0–30 V) across collector–base.

Procedure

14. Set the emitter supply to give $I_E = 2$ mA and hold it fixed for this run.
15. Starting with $V_{CB} = 0$ V, increase V_{CB} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_C at each step.
16. Repeat steps 1–2 for $I_E = 4, 6, 8,$ and 10 mA.
17. Switch off the supply. Plot I_C (Y-axis) vs V_{CB} (X-axis) as a family of curves, one for each I_E .

Observation Table — CB Output Characteristics

S.No.	V_{CB} (V)	I_C (mA) at $I_E=2\text{mA}$	I_C (mA) at $I_E=6\text{mA}$	I_C (mA) at $I_E=10\text{mA}$
1				
2				
3				
4				
5				
6				
7				
8				

7. Part D — CB Configuration: Input Characteristics

Circuit Connections

18. Retain the CB circuit of Part C.

Procedure

19. Set $V_{CB} = 0\text{ V}$ and hold it fixed for this run.
20. Increase V_{EB} in steps of 0.05 V from 0 to 0.4 V , then in steps of 0.02 V up to 0.7 V , recording I_E at each step.
21. Repeat step 2 for $V_{CB} = 5\text{ V}$ and $V_{CB} = 10\text{ V}$.
22. Switch off the supply. Plot I_E (Y-axis) vs V_{EB} (X-axis) as a family of curves, one for each V_{CB} .

Observation Table — CB Input Characteristics

S.No.	V_{EB} (V)	I_E (mA) at $V_{CB}=0\text{V}$	I_E (mA) at $V_{CB}=5\text{V}$	I_E (mA) at $V_{CB}=10\text{V}$
1				
2				
3				
4				
5				
6				
7				
8				

8. Part E — CC Configuration: Output Characteristics

Circuit Connections

23. Connect the transistor in CC configuration: collector grounded (common), base current supplied through a $10\text{ k}\Omega$ resistor from the base supply with a microammeter ($0\text{--}500\text{ }\mu\text{A}$) in series, and emitter connected through a milliammeter ($0\text{--}10\text{ mA}$) to the emitter supply.
24. Connect a voltmeter ($0\text{--}30\text{ V}$) across base–collector and a voltmeter ($0\text{--}30\text{ V}$) across emitter–collector.

Procedure

25. Set the base supply to give $I_B = 20\text{ }\mu\text{A}$ and hold it fixed for this run.
26. Starting with $V_{EC} = 0\text{ V}$, increase V_{EC} in steps of 0.5 V up to 2 V , then in steps of 2 V up to 20 V , recording I_E at each step.
27. Repeat steps 1–2 for $I_B = 40, 60, 80, \text{ and } 100\text{ }\mu\text{A}$.
28. Switch off the supply. Plot I_E (Y-axis) vs V_{EC} (X-axis) as a family of curves, one for each I_B .

Observation Table — CC Output Characteristics

S.No.	V_{EC} (V)	I_E (mA) at $I_B=20\mu\text{A}$	I_E (mA) at $I_B=60\mu\text{A}$	I_E (mA) at $I_B=100\mu\text{A}$
1				
2				
3				
4				

5				
6				
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9. Part F — CC Configuration: Input Characteristics

Circuit Connections

29. Retain the CC circuit of Part E.

Procedure

30. Set $V_{EC} = 1\text{ V}$ and hold it fixed for this run.

31. Increase V_{BC} in steps of 0.5 V from 0 to 10 V , recording I_B at each step.

32. Repeat step 2 for $V_{EC} = 5\text{ V}$ and $V_{EC} = 10\text{ V}$.

33. Switch off the supply. Plot I_B (Y-axis) vs V_{BC} (X-axis) as a family of curves, one for each V_{EC} .

Observation Table — CC Input Characteristics

S.No.	V_{BC} (V)	I_B (μA) at $V_{EC}=1\text{V}$	I_B (μA) at $V_{EC}=5\text{V}$	I_B (μA) at $V_{EC}=10\text{V}$
1				
2				
3				
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10. Calculations / Analysis

- CE: compute $\beta = \Delta I_C / \Delta I_B$ from the output family (at fixed V_{CE}); compute input resistance $r_i = \Delta V_{BE} / \Delta I_B$ and output resistance $r_o = \Delta V_{CE} / \Delta I_C$ from the respective characteristics.
- CB: compute $\alpha = \Delta I_C / \Delta I_E$ from the output family (at fixed V_{CB}); compute $r_i = \Delta V_{EB} / \Delta I_E$ and $r_o = \Delta V_{CB} / \Delta I_C$.
- CC: compute $\gamma = \Delta I_E / \Delta I_B$ from the output family (at fixed V_{EC}); compute $r_i = \Delta V_{BC} / \Delta I_B$ and $r_o = \Delta V_{EC} / \Delta I_E$.
- Cross-check the three current gains using $\beta = \alpha / (1 - \alpha)$ and $\gamma = \beta + 1$, and compare with the directly measured values.
- Comment on the relative magnitudes of r_i and r_o across the three configurations (CE: moderate r_i , moderate r_o ; CB: low r_i , very high r_o ; CC: high r_i , low r_o).